

Lesson plan (template – 2)

Name of the Programme:	B.Sc(H)	
Title of the Course:	Linux and Shell Programming	
Course code	CS1104	
Semester:	I	
Names of the Course Instructor(s)	Prof. Muralidhar Billa Pof. Mytheri S N	
Course credit	No. of hours per week (L + T + P) (2:0:2)	Total no. of Teaching hours
3	4	60

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/ reference
1	1	Overview of Linux: Introduction to Linux distributions	Lecture: Instructor explains Linux distributions.	Wale Soyinka, Ch. 1
2		Basic architecture of Linux	Lecture: Explain Linux kernel and shell. Tutorial: Students label a Linux architecture diagram and classify 3 components (e.g., file system, processes) in groups.	Wale Soyinka, Ch. 1
3		Lab Program 1 - Create, copy files (ls, cp, mkdir)	Students create 3 directories, copy 2 text files using cp, and list contents with ls in a terminal, submitting a screenshot of commands and output.	James Arthur, Ch. 2
4		Lab Program 1 - Move, delete files (mv, rm, rmdir)	Students move files with mv, delete 2 files using rm, remove a directory with rmdir, and verify changes with ls, submitting a lab log.	James Arthur, Ch. 2
5		Basic Commands: File management (ls, cp, mv)	Lecture: Instructor demos ls, cp, mv. Interactive Discussion: Students solve 3 file management scenarios (e.g., copy directory) in small groups.	Wale Soyinka, Ch. 2
6		Viewing file content (cat, more, less)	Demo: (cat, more, less) Interactive Discussion: Students solve 3 file management scenarios (e.g., cat directory) in small groups.	Wale Soyinka, Ch. 2

7		Lab Program 2 - Investigate/modify file permissions (chmod)	Students change permissions using chmod (e.g., 644 to 755), verify with ls -l, and explain changes in a pair discussion, submitting a log.	James Arthur, Ch. 3
8		Lab Program 2 - Investigate permissions for users/groups (chown, chgrp)	Students modify permissions for users/groups using chmod and chgrp, test access with a test user, and submit a report with results.	James Arthur, Ch. 3
9		File permissions: chmod, chown	Lecture: Explain chmod and chown. Students solve 3 permission scenarios (e.g., restrict access) using chmod and chown in a guided exercise.	Wale Soyinka, Ch. 3
10		Ownership commands (chown, chgrp)	Demo: Instructor shows chown and chgrp on sample files. Interactive Discussion: Students debate user vs. group ownership for 2 scenarios, presenting findings.	Wale Soyinka, Ch. 3
11		Lab Program 3 - Display file content with line numbers (cat -n)	Students use cat -n and nl on a text file, compare outputs, and write a script to display numbered content, submitting the script and output.	Sumitabha Das, Ch. 5
12		Lab Program 3 - Explore file content commands (more, less)	Students navigate a large file using more and less, search for specific text, and document navigation steps in a lab report.	Sumitabha Das, Ch. 3
13	2	Introduction to shell types (Bash, Zsh)	Lecture: Explain Bash and Zsh features. Interactive Discussion: Students compare Bash and Zsh in groups, listing 2 pros and cons for each on a shared doc.	Wale Soyinka, Ch. 5
14		Creating and executing simple shell scripts	Demo: Instructor creates a script to print "Hello". Students write and execute a similar script, debug errors, and submit the script file.	Wale Soyinka, Ch. 5
15		Lab Program 4 - Count words,	Hands-on Lab: Students use wc to count words and lines, write a	James Arthur, Ch. 7

		lines in a file (wc)	script to automate counting for 2 files, and submit script and output.	
16		Lab Program 4 - Count characters in a file	Hands-on Lab: Students use wc -m to count characters, test on 2 files, and submit a lab report with character counts and commands.	James Arthur, Ch. 7
17		Variables: Defining and using variables	Lecture: Explain variable syntax. Tutorial: Students define 3 variables in a script, print values, and solve 2 variable-based exercises in pairs.	Randal K. Michael, Ch. 2
18		Environment vs. local variables, special variables	Demo: Instructor shows export and special variables (\$?, \$#). Interactive Discussion: Students analyze outputs of 3 special variables in groups.	Randal K. Michael, Ch. 2
19		Lab Program 5 - Arithmetic operations: addition, subtraction	Hands-on Lab: Students write a script for addition and subtraction with user input, test with 3 inputs, and submit the script and output screenshots.	James Arthur, Ch. 4
20		Lab Program 5 - Arithmetic operations: multiplication, division	Hands-on Lab: Students extend the script for multiplication and division, handle division by zero, and submit updated script with test results.	James Arthur, Ch. 4
21		Input/Output: Using echo for output	Lecture: Explain echo syntax. Tutorial: Students write 2 scripts using echo with variables, format outputs, and present one to the class.	Wale Soyinka, Ch. 6
22		Reading user input with read command	Demo: Instructor shows read in a script. Tutorial: Students create a script using read, validate input, and solve 2 input-related problems.	Wale Soyinka, Ch. 6
23		Lab Program 6 - String concatenation	Hands-on Lab: Students write a script to concatenate 2 user-input strings, test with 3 cases, and submit the script and concatenated outputs.	Sumitabha Das, Ch. 6

24		Lab Program 6 - Substring extraction	Hands-on Lab: Students extract substrings using cut or parameter expansion, test 3 cases, and submit a lab report with results and commands used.	Sumitabha Das, Ch. 6
25	3	Conditional statements: if, else	Lecture: Explain if-else syntax. Tutorial: Students draw flowcharts for 2 conditions, write an if-else script, and test in pairs.	Randal K. Michael, Ch. 3
26		Conditional statements: elif, case	Demo: Instructor shows elif and case. Interactive Discussion: Students solve a grade calculator scenario using case, presenting solutions.	Randal K. Michael, Ch. 3
27		Lab Program 7 - Generate multiplication table	Hands-on Lab: Students write a script to print a multiplication table for a user-input number, test with 3 inputs, and submit the script and output.	James Arthur, Ch. 5
28		Lab Program 7 - Debug multiplication table script	Hands-on Lab: Students debug a faulty multiplication table script, fix errors, and submit a lab report explaining corrections and final output.	James Arthur, Ch. 5
29		Looping constructs: for loop	Lecture: Explain for loop syntax. Tutorial: Students write a for loop script to iterate over files, solve 2 loop problems in groups.	Wale Soyinka, Ch. 7
30		Looping constructs: while loop	Demo: Instructor shows while loop for counting. Tutorial: Students write a while loop script, test 3 cases, and share outputs.	Wale Soyinka, Ch. 7
31		Lab Program 8 - Remove lines with specific word (grep)	Hands-on Lab: Students use grep -v to exclude lines with a word, test on 2 files, and submit filtered outputs with commands used.	Sumitabha Das, Ch. 8
32		Lab Program 8 - Remove lines using sed	Hands-on Lab: Students use sed to delete lines with a pattern, test on 2 files, and submit a lab report with modified files and commands.	Sumitabha Das, Ch. 8

33	4	Text Processing: Introduction to grep	Lecture: Explain grep syntax. Demo: Instructor filters text with grep. Tutorial: Students filter 3 patterns in a file using grep.	Wale Soyinka, Ch. 9
34		Text Processing: Introduction to sed	Lecture: Explain sed basics. Demo: Instructor replaces text with sed. Tutorial: Students write 2 sed commands to modify a file.	Wale Soyinka, Ch. 9
35		Lab Program 8 - Advanced regex with grep	Hands-on Lab: Students use regex with grep to filter complex patterns, test on 2 files, and submit filtered outputs with commands used.	Sumitabha Das, Ch. 9
36		Lab Program 8 - Regex with sed	Hands-on Lab: Students use sed with regex to replace patterns, test on 2 files, and submit a lab report with modified files and commands.	Sumitabha Das, Ch. 9
37		Text manipulation with awk	Lecture: Explain awk syntax. Demo: Instructor extracts columns with awk. Tutorial: Students extract 2 columns from a file using awk.	Randal K. Michael, Ch. 5
38		Advanced awk for text processing	Lecture: Explain advanced awk (e.g., calculations). Tutorial: Students use awk to sum a column, solve 2 problems, and share results.	Randal K. Michael, Ch. 5
39		Lab Program 9 - List files with permissions (ls -l)	Hands-on Lab: Students run ls -l, interpret permission strings for 5 files, and classify permissions (e.g., read-only) in a lab report.	Sumitabha Das, Ch. 3
40		Lab Program 9 - Advanced file listing with permissions	Hands-on Lab: Students use ls with options (-a, -R) to list hidden files and directories, analyze permissions, and submit a detailed permission analysis.	Sumitabha Das, Ch. 3
41		Regular Expressions: Basics	Lecture: Explain regex syntax. Interactive Discussion: Students match 3 regex patterns in a worksheet, discussing results in groups.	Wale Soyinka, Ch. 10

42		Using regex with grep	Demo: Instructor combines grep with regex. Tutorial: Students write 3 regex-based grep commands, test on a file, and share outputs.	Wale Soyinka, Ch. 10
43		Lab Program 10 - Compare two files (diff command)	Hands-on Lab: Students use diff to compare 2 files, analyze differences, and submit a report with comparison results and command output.	James Arthur, Ch. 6
44		Lab Program 10 - Advanced file comparison	Hands-on Lab: Students use cmp and comm to compare files, test 3 file pairs, and submit a lab log with differences and commands used.	James Arthur, Ch. 6
45	5	Process Basics: ps command, internal/external commands	Lecture: Explain ps and command types. Tutorial: Students run ps, classify 5 commands as internal/external, and submit a worksheet.	Wale Soyinka, Ch. 11
46		Running jobs in the background	Demo: Instructor runs a command with &. Tutorial: Students run 2 commands in the background, monitor with jobs, and document behavior.	Wale Soyinka, Ch. 11
47		Lab Program 11 - Display system info: date, users	Hands-on Lab: Students use date and who, write a script to combine outputs, test, and submit the script with output screenshots.	James Arthur, Ch. 8
48		Lab Program 11 - Display system uptime	Hands-on Lab: Students use uptime, parse output in a script, and submit a lab report with uptime details and script used.	James Arthur, Ch. 8
49		Process management: nice, at commands	Lecture: Explain nice and at. Tutorial: Students adjust priorities with nice, schedule a task with at, and solve 2 scenarios.	Randal K. Michael, Ch. 6
50		Process management: batch, cron	Demo: Instructor schedules a task with cron. Tutorial: Students write 2 crontab entries, test scheduling, and discuss results.	Randal K. Michael, Ch. 6

51		Lab Program 11 - Schedule tasks with cron	Hands-on Lab: Students create a crontab entry to run a script, test scheduling, and submit the crontab file and script output.	Sumitabha Das, Ch. 10
52		Lab Program 11 - Advanced cron scheduling	Hands-on Lab: Students modify crontab for complex scheduling (e.g., every 5 minutes), test, and submit a lab report with crontab and results.	Sumitabha Das, Ch. 10
53		Customizing Environment: System variables, profile	Lecture: Explain system variables and .bashrc. Tutorial: Students set a variable in .bashrc, test, and document changes.	Wale Soyinka, Ch. 12
54		Aliases, command history	Demo: Instructor creates an alias and uses history. Tutorial: Students create 2 aliases, explore history, and submit commands used.	Wale Soyinka, Ch. 12
55		Lab Program 11 - Create aliases	Hands-on Lab: Students create 3 aliases (e.g., ll for ls -l), test in a script, and submit a lab report with alias definitions and outputs.	James Arthur, Ch. 9
56		Lab Program 11 - Explore command history	Hands-on Lab: Students use history to recall commands, save 5 commands to a file, and submit the file with a report on history usage.	James Arthur, Ch. 9
57		Review of process management and customization	Lecture: Recap cron, aliases, and variables. Interactive Discussion: Students solve 3 process management scenarios in groups, presenting solutions.	Wale Soyinka, Ch. 11-12
58		Course revision and CIE preparation	Lecture: Summarize key concepts. Tutorial: Students complete a practice quiz on modules 1-5 and discuss answers in pairs.	Sumitabha Das, Ch. 1-10
59		Lab: End to end System Administration Task	Hands-on Lab: Students use all program knowledge and do commands to a file, and submit the file along with the automation Job.	James Arthur, Ch. 2-5

		Automation Script		
60		Lab: End to end IT Support Task Automation Script	Hands-on Lab: Students use all program knowledge and do commands to a file, and submit the file along with the automation Job.	James Arthur, Ch. 4-7

Note: 1

2 credit courses	30 hours
3 credit courses	45 hours
4 credit courses	60 hours

Note: 2

- A course shall have a minimum of 3 modules and maximum 5 modules
- Each module will be covered in minimum 6 hours and maximum 12 hours

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1	
- IBM SkillsBuild Courses: Graded Component-1	10 marks out of 70
- Quiz Questions Graded Component 2	10 marks out of 70
CIE-2	25 marks out of 70
- Theory	
CIE-3	
- Lab record writeup & viva	5 marks out of 70
- Lab Practical Test	10 marks out of 70
- Lab Test Viva	10 marks out of 70

Semester End Exams (SEE)	
Mode of Exam	Practical
Weightage of exam	30

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	CO1: Explore the architecture of Linux and its distributions to perform effective file, resource, and process management.	3	2	2	3	1	-	1	2	-	-	-	-
CO2	CO2: Apply essential Linux commands for file operations, permission handling, and environment customization.	3	2	1	3	1	-	1	2	-	1	-	-
CO3	CO3: Apply basic shell scripts using variables, control structures, and functions to automate tasks and enhance script functionality.	3	3	2	3	1	1	2	2	-	-	-	-
CO4	CO4: Perform the text processing tools and regular expressions for content manipulation and system scripting.	3	3	2	3	1	-	1	2	-	-	-	-

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	IBM SkillsBuild Courses: Graded Component-1	Certificate of completion for 10 marks. The specified courses on the IBM SkillsBuild platform to be uploaded to the designated Google Classroom. (Refer Table-1)
	Quiz Questions Graded Component 2	Quiz questions will be conducted for 10 marks. Total 20 Questions will be given each right answer carries half mark each. No negative marking for wrong answers.

Table-1:

Graded Component-1 (10 Marks):

Assess students' ability to complete assigned courses on the IBM SkillsBuild platform and reflect on their learning.

Criteria	Excellent (10 Marks)	Satisfactory (5 Marks)	Needs Improvement (0 Marks)
Completion of Courses (10 Marks)	Successfully completes all specified courses on the IBM SkillsBuild platform before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Graded Component-2 (10 Marks):

Criteria	Correct answer (0.5 Marks)	Wrong Answer (0 Marks)
Quiz questions (10 Marks)	Successfully choosing the correct option Student will be awarded (0.5 marks).	Fails to choose correct option will be reward as (0 marks).

Note: Total 20 Questions will be given each right answer carries 0.5 mark each. No negative marking for wrong answers

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Lab record+ Viva = 5 marks (<i>Refer Table-2</i>) Lab Practical Test = 10 marks (<i>Refer Table-3</i>) Lab Test Viva = 10 marks (<i>Refer Table-3</i>)

Table-2

Lab Write-up and Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1	Understanding of problem statements. (2 Marks)	Observations	Student exhibits thorough understanding of requirements and applies suitable techniques for the problem. (2 M)	Students have sufficient understanding of requirements and apply suitable techniques for the problem. (1 M)	Student does not have a clear understanding of requirements and is unable to apply any techniques for the problem. (0 M)
2	Execution (2 Marks)	Observations	Students demonstrate the execution of the program with error correction and show performance efficiency. Appropriate validations with all test cases are handled. (2 M)	Students demonstrate the execution of the program without error correction of the code and show performance efficiency with only few test cases. (1 M)	Student has not executed the program. (0 M)
3	Format of Results (2 Marks)	Observations	Documentation with appropriate comments and output is covered in data sheets and manual. (2 M)	Documentation with only few comments and only few output cases is covered in data sheets and manual. (1 M)	Documentation with no comments and no output cases is covered in data sheets and manual. (0 M)
Viva Voce Rubrics (Max: 4 marks)					

1	Conceptual Understanding (2 Marks)	Viva Voce	Explains thoroughly the program executed along with related concepts. (2 M)	Adequately explains the program executed along with related concepts (1 M)	Unable to explain the program executed along with related concepts (0 M)
2	Communication of Concepts (2 Marks)	Viva Voce	Communicates the concept used in problem solving well. (2 M)	Sufficiently communicates the concepts used in problem solving. (1 M)	Unable to communicate the concepts used in the problem. (0 M)
TOTAL MARKS OBTAINED (10)					

Note: Overall Lab Record including 11 programs where Write-up: 6 Marks & Viva-Voce: 4 Marks will sum up to 10 M, will be reduced to 5 Marks.

Table-3

Lab Test rubrics (Max:10 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1	Implementation Writing (5 Marks)	Observations	Well-organized, clear, with proper logic of the problem. validations with all test cases are handled. (5 M)	Well-organized, clear, with partial logic of the problem and validations. (1 - 4 M)	Student has not executed the program. (0 M)
2	Execution and Format of Results (5 Marks)	Observations	Students demonstrate the execution of the program with error correction and show performance efficiency. Appropriate validations with test cases (minimum 2) are	Students demonstrate the execution of the program without error correction and appropriate validations	Execution Results with no comments and no output (0 M)

			handled. Execution Results should show in proper format (5 M)	with test case (minimum 1) is handled. Execution Results in without format (1- 4 M)	
Viva Voce Rubrics (Max: 10 marks)					
1	Conceptual Understanding (5 Marks)	Viva Voce	Explains thoroughly the program executed along with related concepts. (5 M)	Adequately explains the program executed along with related concepts (3 – 4 M)	Unable to explain the program executed along with related concepts (1-2 M)
2	Communicatio n of Concepts (5 Marks)	Viva Voce	Communicates the concept used in problem solving well. (5 M)	Sufficiently communicat es the concepts used in problem solving. (3-4 M)	Unable to commun icate the concepts used in the problem. (1-2 M)

Note: Overall Lab test will be conducted including 2 programs where Write-up: 10 Marks & Viva-Voce: 5 Marks will sum up to 30 M, will be reduced to 20 Marks.

Signature of the Course Faculty